# Sazid Ali Rangrej

bm23btech11020@iith.ac.in | sazidalir@gmail.com | +91 8955662539 | LinkedIn | GitHub

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.84 |
| ZEE SCHOOL PRAGATI PATH BHILWARA | Class XII (CENTRAL BOARD OF SECONDARY EDUCATION) | 2023 | 90.00% |
| KENDRIYA VIDYALAYA BHILWARA | Class X (CENTRAL BOARD OF SECONDARY EDUCATION) | 2020 | 90.80% |

## EXPERIENCE

**AI/ML Research Intern** | MUFG Global Service

*with MUFG Securities EMEA plc*

- Developed **Ground Truth by Construction (GTbC)**, a novel QA-system evaluation method that is both **reliable and scalable**, unlike expert annotation or LLM-generated ground truth (RAGAS, ARES, DeepEval).
- Raised evaluation difficulty **beyond the evaluator's own capability ceiling** to enable testing of next-generation QA systems, proven when the generator model (Opus 4.7) scored under 40% on its own documents.
- Evaluated three QA systems built on Claude models (Haiku 4.5, Sonnet 4.6, Opus 4.7) on 50 questions from a Basel III regulatory document. Accuracy collapsed from **98-100% to 14-40%,** with the strongest model ranking worst on ambiguity.
- Co-authored the resulting research paper, published on SSRN and currently under review at **ACL Rolling Review (ARR)**.

**Teaching Assistant** | Data Structures and Algorithms Lab, IIT Hyderabad

- Assisted Prof. Nagarajan Ganapathy in teaching an undergraduate lab course on Data Structures and Algorithms.
- Conducted weekly tutorials to resolve students' doubts and reinforce lab concepts.
- Curated problem sets for weekly assignments and hackathons.

## PROJECTS

**Neural Hangman Solver |** Meltwater Hackathon · PyTorch, Transformers, BiGRU, Beam Search

- Built a two-model system (4.1M-param BiGRU + 6-layer Transformer LM) in PyTorch, scoring **71.59%** win rate on 250,000 unseen words with zero train/test vocabulary overlap, vs 14.6% frequency baseline.
- Generated reachable game states calibrated to real play, then fine-tuned on the model's own boards (self-play) to fix distribution shift (**+2.85pp**).
- Routed between models by hidden-position count based on beam-search coverage (**+2.6pp held-out**); batched, length-bucketed GPU beam search (**6.8x faster**), verified against a reference implementation.

**Intelligent Email Automation Agent | LangGraph, Phi-3-mini, LoRA/PEFT, Gmail API**

- Fine-tuned a quantised **Phi-3-mini (3.8B)** with LoRA for 3-class email intent classification, reaching **weighted F1 0.70** and **80% end-to-end task success** on a held-out set of 50 emails.
- Built a perceive-think-act LangGraph workflow that extracts deadlines, creates Google Calendar events, and drafts context-aware replies over unread mail.
- Added human-in-the-loop approval before any send action, with logged reasoning traces per decision.

**LLM-Powered Retrieval & Re-Ranking for Video Search** | Motive Hackathon · AWS Bedrock, FAISS, RAG

- Built a hybrid retrieval pipeline for video search, combining sparse TF-IDF with dense AWS Bedrock Titan embeddings in FAISS, plus query normalisation (lowercasing, stop-word removal, token cleanup).
- Added an LLM re-ranker over the top-k candidates that returns a relevance score together with a short justification, rather than a bare similarity value.

## SKILLS

- **Fundamentals:** Data Structures, Algorithms, OOP
- **Languages:** Python, C++, C
- **LLM & Agentic AI:** LangGraph, LangChain, RAG, LoRA fine-tuning
- **ML & Retrieval:** PyTorch, Hugging Face Transformers, FAISS, embeddings, semantic search, AWS Bedrock
- **Backend & Deployment:** FastAPI, REST APIs, Docker, CI/CD (GitHub Actions)
- **Tools:** Git, GitHub, Jupyter

## RELEVANT COURSES

- **Computer Science/AI:** Algorithms & Data Structure Lab, Introduction to Programming, Artificial Intelligence.
- **Mathematics:** Linear Algebra, Statistics, Probability and Random Processes, Calculus, Introduction to Probability.

## POSITIONS OF RESPONSIBILITY

**Consulting Coordinator | Finance and Consulting Club,** IIT Hyderabad

- Co-led the planning and execution of HORIZON, the club's flagship event, hosting 20+ industry speakers and 12+ competitions across consulting and business domains.

**Core Team Member | Behind the Lens,** IIT Hyderabad

- Contributed to scripting, direction, and production planning for the club's short films.

## EXTRACURRICULAR

- Selected to represent IIT Hyderabad in the filmmaking contingent at Inter-IIT Cultural Meet 7.0 (2024), IIT Patna.
- Institute-level finalist in Smart India Hackathon (SIH) 2024
- NSS (National Service Scheme) volunteer, engaged in rural teaching and social awareness initiatives.